PYL1002: Waves and Oscillations **Tutorial - 2**
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Collaborative Problem Solving on Simple Harmonic Oscillation

1 The Interatomic "Morse" Potential [15 points]

A simple, albeit useful model for interatomic interactions of diatomic molecules is given by the *Morse* Potential. Consider the problem in one-dimension, with x being the separation between the two atoms, for which the Morse potential energy takes the following form (where D, a, b are positive constants),

$$U(x) = D\left(1 - e^{-a(x-b)}\right)^2 . \tag{1}$$

a What are the units of D, a and b, respectively?

b Find the separation $x = x_0$ for which the net forces on the atoms vanish, *i.e.*, the system is in equilibrium. Is this equilibrium stable or unstable?

c Now expand the Morse potential $U(x)$ around the equilibrium position x_0 to approximate it as a harmonic potential.

d For small oscillations of the interatomic bond around the equilibrium position $x = x_0$, what is the frequency of this oscillation in terms of the given variables? Schematically plot the Morse potential and mark relevant quantities such as equilibrium length, curvature at equilibrium, etc.

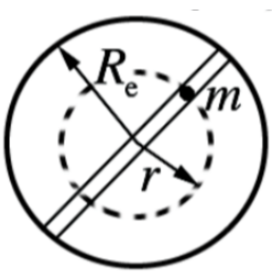
I want to now point your attention to another interatomic potential, the Lennard-Jones (LJ) potential we discussed during lecture last week, where α and β are positive constants,

$$U_{LJ}(x) = -\frac{\alpha}{x^6} + \frac{\beta}{x^{12}} . \tag{2}$$

e For what constraint equation(s) connecting α and β to a, b and D from the Morse potential above, would give the same vibrational frequency of the molecule?

2 Falling Through the Earth = Orbiting The Earth?

Imagine that one drilled a hole from Delhi straight through the center of the earth to the other end which happens to be a point in the South Pacific Ocean, somewhere close to Chile's Easter Island[1]. We want to understand how a particle dropped into this extremely interesting tunnel would behave. Let's take the mass of the Earth to be M and radius R_e.



[1]This is called the Antipode of a given point on Earth, check out https://www.antipodesmap.com/.

a First things first, let's find the force on a particle of mass m dropped into the tunnel. As you may recall from your earlier exposition of Newton's Law of Gravity, the gravitational force on this mass at a distance $r < R_e$ from the center of the Earth is due to only the mass of the earth that lies within[2] a solid sphere of radius r. Using this, and assuming the Earth has a uniform mass density, find the force on the particle at distance r. The universal gravitation constant is G.

b You would notice that the force you found above looks like simple harmonic motion! Write an expression for the time period for one rount trip in terms of M and R_e. To get a sense of this time scale, find the time taken for a particle dropped from Delhi to reach to its antipodal point through this tunnel. You can take $M = 6 \times 10^{24}$ kg, and $R_e = 6 \times 10^3$ km.

c Next, I want you to watch this YouTube short featuring Neil deGrasse Tyson: https://www.youtube.com/shorts/7_VxUZQwM80. Here, Neil claims that the time period of falling through a tunnel connecting ***any two points*** (not necessarily through the center) on the Earth is the same. Prove this!

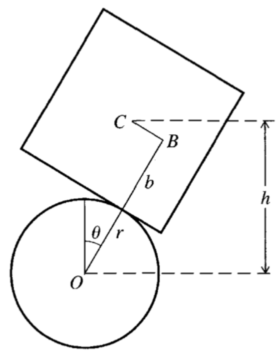
d Now, for a final twist. Imagine a satellite orbiting the Earth in a circular orbit not too far from the Earth's surface. In terms of M and R_e, calculate the time taken for this satellite to cover one revolution around the Earth. How does it compare with the time taken to fall through the Earth?

e Understand this connection by looking at the projection of circular motion onto a diameter of the circle. This should also give you an appreciation why the angular frequency ω turns up in both simple harmonic and circular motion.

As a final wrap up to this problem, I encourage that **after class,** you check out this YouTube video by Prof. Singh on the same topic which should hopefully reflect the solution you have developed: https://www.youtube.com/watch?v=TI9biu93i4w

3 A Balancing Act

In an effort to impress your young nephews and nieces, you come with a trick. You wish to balance a wooden cube of side $2b$ on a hard rubber cylinder of radius r which is held fixed with its axis horizontal. The four sides of the cube are parallel to the cylinder axis. The cube cannot slip on the rubber of the cylinder, but it can of course rock from side to side, as shown in the figure. To ensure your trick works, you need to understand the physics behind when the cube would rock or when it would fall.



a Convince yourself that the angle θ is enough to characterize the configuration of the system fully. Write down an expression for the potential energy $U(\theta)$. It's just gravitational potential energy after all. To help you get started, try writing h as a function of θ using geometry. You may find it helpful to know that since the cube cannot slip, the length BC must be equal to $r\theta$. Think about it.

b As you may imagine, $\theta = 0$ is a point of local equilibrium. Prove this using energy considerations. Remember $F(\theta) \propto -dU(\theta)/d\theta$

c Finally, for small deviations of θ around $\theta = 0$, there are two possibilities. The cube may rock back-and-forth, or it may get unstable and fall off. Under what condition would the equilibrium be stable and the cube would oscillate? Think second derivative of the potential energy. Does the result make intuitive sense?

[2]One can also understand this from a gravitational analog of Gauss' Law in electromagnetism. Just wanted to mention this if you wanted to further appreciate this connection.